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# Image Sensing and Acquisition in Low-Light Conditions

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload an image
uploaded = files.upload()

# Read the uploaded image
image_name = list(uploaded.keys())[0]
img = cv2.imread(image_name)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Simulate low-light image
low_light = (img * 0.3).astype(np.uint8)

# Add sensor noise
noise = np.random.normal(0, 20, low_light.shape).astype(np.uint8)
noisy_image = cv2.add(low_light, noise)

# Enhance the image
enhanced = cv2.convertScaleAbs(noisy_image, alpha=2.0, beta=30)

# Display results
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(noisy_image)
plt.title("Low-Light Captured Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(enhanced)
plt.title("Enhanced Image")
plt.axis("off")

plt.show()
```

[Choose Files](#) LION.jpg

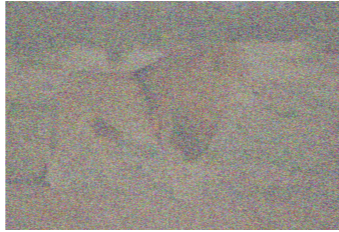
LION.jpg(image/jpeg) - 143625 bytes, last modified: 7/27/2026 - 100% done

Saving LION.jpg to LION.jpg

Original Image



Low-Light Captured Image



Enhanced Image

